Chapter1: THE FOUNDATIONS OF LOGIC AND PROOFS

1.1 PROPOSITIONAL LOGIC:

Proportions:

is a statement that is either true or false, but not both.

Example of proportions:

1- Washington is the capital of USA.
2- Toronto is the capital of Canada.
3- 1 + 1 = 2.
4- 2 + 2 = 3.

Some Sentences are not proportions:

1- What time is it?
2- Read this carefully.
3- x + 1 = 2.
4- x + y = z.

Logical Operators:

1- Negation of p

Let p be a proposition. The negation of p, denoted by ($\sim p$ or $\bar{p}$), is the statement is the opposite of the truth value of p.

p	$\neg p$
T	F
F	T

2- Conjunction of p and q

Let p and q be propositions. The conjunction of p and q, denoted by (p ∧ q) is true when both p and q are true and is false otherwise.

3- Disjunction of p and q

Let p and q be propositions. The disjunction of p and q, denoted by (p ∨ q) is false when both p and q are false and is true otherwise.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

4- Exclusive or of p and q

Let p and q be propositions. The exclusive or of p and q, denoted by $(p \oplus q)$ is true when exactly one of p and q is true and is false otherwise.

5- Conditional of p and q

Let p and q be propositions. The conditional statement "if p, then q" denoted by $(p \rightarrow q)$ is false when p is true and q is false, and true otherwise.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

6- Biconditional of p and q

Let p and q be propositions. The biconditional statement "p if and only if q" denoted by $(p \leftrightarrow q)$ is true when p and q have the same truth values, and is false otherwise.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Compound proportions:

Ex: Construct the truth table of the compound proportion

$$(p \vee \sim q) \rightarrow (p \wedge q)$$

Sol:

p	q	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \rightarrow (p \wedge q)$
T	T	F	T	T	T
T	F	T	T	F	F
F	T	F	F	F	T
F	F	T	T	F	F

Precedence of Logical Operators:

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

1.2 PROPORTIONAL EQUIVALENCES

Tautology:

A compound proposition that is always true, no matter what the truth values of the propositions is called a **tautology**.

Contradiction:

A compound proposition that is always false is called a **contradiction**.

Ex: Determine whether the following propositions are **tautology** , **contradiction** or non:

1- $(p \vee \sim p)$ 2- $(p \wedge \sim p)$

Sol:

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

$\therefore$ $(p \vee \sim p)$ is tautology

$\therefore$ $(p \wedge \sim p)$ is contradiction

Logical Equivalences:

The compound propositions p and q are **logically equivalent** if p ↔ q is a tautology (or if p and q have the same truth values) , and denoted by (p ≡ q).

Ex: Prove the following propositions are logically equivalent.

1) $\sim(p \vee q)$ and $\sim p \wedge \sim q$ 2) $p \rightarrow q$ and $\sim p \vee q$

Sol:

1) $\sim(p \vee q)$ and $\sim p \wedge \sim q$

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

≡

∴ ~(p ∨ q) ≡ ~p ∧ ~q

2) $p \rightarrow q$ and $\sim p \vee q$

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

≡

∴ $p \rightarrow q \equiv \sim p \vee q$

TABLE 5 Logical Equivalences.

Equivalence	*Name*
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

TABLE 6 Logical Equivalences Involving Conditional Statements.

$p \rightarrow q \equiv \neg p \vee q$
$p \rightarrow q \equiv \neg q \rightarrow \neg p$
$p \vee q \equiv \neg p \rightarrow q$
$p \wedge q \equiv \neg(p \rightarrow \neg q)$
$\neg(p \rightarrow q) \equiv p \wedge \neg q$
$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$
$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$
$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$
$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

TABLE 7 Logical Equivalences Involving Biconditionals.

$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$
$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

Ex: Prove the following without truth table:

1) ~ (p → q) and $(p \wedge \sim q)$ is logically equivalent.
2) $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

Sol:

1) $\neg(p \rightarrow q) \equiv \neg(\neg p \vee q)$ by Example 3
$\equiv \neg(\neg p) \wedge \neg q$ by the second De Morgan law
$\equiv p \wedge \neg q$ by the double negation law

2) $(p \wedge q) \rightarrow (p \vee q) \equiv \neg(p \wedge q) \vee (p \vee q)$ by Example 3
$\equiv (\neg p \vee \neg q) \vee (p \vee q)$ by the first De Morgan law
$\equiv (\neg p \vee p) \vee (\neg q \vee q)$ by the associative and commutative laws for disjunction
$\equiv \mathbf{T} \vee \mathbf{T}$ by Example 1 and the commutative law for disjunction
$\equiv \mathbf{T}$ by the domination law

1.3 PREDICATES AND QUANTIFIERS:

Predicates:

Statements involving variables, such as

"x > 3," "x = y + 3," "x + y = z,"

Ex: What are the truth values of the following:

1- P(x) : x > 3 , if P(4) and P(2)?
2- Q(x, y) : x = y + 3 , if Q(1, 2) and Q(3, 0)?

Sol:

1- P(4) : 4 > 3 , is true
P (2): 2 > 3 , is false.

2- Q(1, 2): 1 = 2 + 3 , is false
Q(3, 0) : 3 = 0 + 3 , is true.

Quantifiers

1- Universal Quantification:

The universal quantification of P(x) is the statement "P(x) for all values of x in the domain". The notation $\forall x\ P(x)$.

2- Existential Quantification:

The existential quantification of P(x) is the statement "There exists an element x in the domain such that P(x)". The notation $\exists x\ P(x)$.

Ex: What is the truth value of the following:

1- $\forall x\ (x + 1 > x)$, where x is all real numbers.
2- $\forall x\ (x < 2)$, where x is all real numbers.
3- $\exists x\ (x > 3)$, where x is all real numbers.
4- $\exists x\ (x=x+1)$, where x is all real numbers.

Sol:

1- $\because (x + 1 > x)$ is true for all real numbers x
$\therefore \forall x\ (x + 1 > x)$ is true

2- $\because (x < 2)$ is not true for all real number x, for example when x=3 is false.
$\therefore \forall x\ (x < 2)$ is false

3- $\because (x > 3)$ is sometimes true , for example when x=4
$\therefore \exists x\ (x > 3)$ is true.

4- $\because (x=x+1)$ is false for all real number x
$\therefore \exists x\ (x=x+1)$ is false.

1.4 PROOFS:

Direct Proof

Direct proofs start with the hypothesis and end with the conclusion.

Definition1:

1- Even Integer:

The integer n is even if $n = 2k \ , \ k \in \mathbb{Z}$.

2- Odd Integer:

The integer n is odd $if \ \ n = 2k + 1 \ , \ k \in \mathbb{Z}$.

Note: The integer is either even or odd, and no integer is both even and odd.

Ex: Prove:

1- If n is an odd integer, then n^2 is odd.
2- If n is an even integer, then n^2 is even.

Sol:

1- Let n is an odd integer $\Rightarrow \quad n = 2k + 1 \ , \ k \in Z$

$$\Rightarrow \quad (n)^2 = (2k + 1)^2$$

$$\Rightarrow \quad n^2 = 4k^2 + 4k + 1$$

$$\Rightarrow \quad n^2 = 2(2k^2 + 2k) + 1$$

$$\Rightarrow \quad n^2 = 2m + 1 \ , \quad m = 2k^2 + 2k$$

$\therefore \ n^2$ is odd integer

2- Let n is an even integer $\Rightarrow \quad n = 2k \ , \ k \in Z$

$$\Rightarrow \quad (n)^2 = (2k)^2$$

$$\Rightarrow \quad n^2 = 4k^2$$

$$\Rightarrow \quad n^2 = 2(2k^2)$$

$$\Rightarrow \quad n^2 = 2m \ , m = 2k^2$$

$\therefore \ n^2$ is even integer

Indirect proof

Proofs of theorems that are not direct proofs, that is, that do not start with the hypothesis and end with the conclusion, are called indirect proofs.

An extremely useful two types of indirect proof are known as proof by **contraposition** and proof by **contradiction**.

1- Proof by Contraposition

The contrapositive of a conditional statement $(p \to q)$ is $(\sim q \to \sim p)$

$$p \to q \equiv \sim q \to \sim p$$

Ex: Prove the following by contraposition:

a) If n is an integer and 3n + 2 is odd, then n is odd.
b) If n^2 is an odd integer, then n is odd.

Sol:

a) We will Prove " If n is an integer and n is even, then 3n + 2 is even".

Let n is even $\Rightarrow \quad n = 2k \ , \ k \in Z$

$$\Rightarrow \quad 3n + 2 = 3(2k) + 2$$

$$\Rightarrow \quad = 6k + 2$$

$$\Rightarrow \quad = 2(3k + 1)$$

$$\Rightarrow \quad = 2(3k + 1) \ , \ m = 3k + 1$$

$\therefore \ 3n + 2$ is even integer

b) We will Prove " If n is n is even, then n^2 is even".

Let n is even $\Rightarrow \quad n = 2k \ , \ k \in Z$

$$\Rightarrow \quad (n)^2 = (2k)^2$$

$$\Rightarrow \quad n^2 = 4k^2$$

$$\Rightarrow \quad n^2 = 2(2k^2)$$

$$\Rightarrow \quad n^2 = 2m \ , m = 2k^2$$

$\therefore \ n^2$ is even integer

2- Proofs by Contradiction

To prove the conditional statement $(p \rightarrow q)$ by contradiction, we assume the negation of the conclusion $(\sim q)$.

Assume $(p$ and $\sim q)$

Ex: Prove the following by contradiction:

a) If n is an integer and 3n + 2 is odd, then n is odd.
b) If n^2 is an odd integer, then n is odd.

Sol:

a) Let 3n + 2 is odd and n is even

$$\therefore n = 2k \ , \ k \in Z$$

$$\therefore 3n+2 = 3(2k)+2$$

$$\Rightarrow \quad = 6k+2$$

$$\Rightarrow \quad = 2(3k+1)$$

$$\Rightarrow \quad = 2m \ , \ m = 3k+1$$

$\therefore 3n+2$ is even integer, but this is contradiction.

So " if 3n + 2 is odd ,then n is odd"

b) Let n^2 is odd integer and n is even

$$\Rightarrow \quad n = 2k \ , \ k \in Z$$

$$\Rightarrow \quad (n)^2 = (2k)^2$$

$$\Rightarrow \quad n^2 = 4k^2$$

$$\Rightarrow \quad n^2 = 2(2k^2)$$

$$\Rightarrow \quad n^2 = 2m \ , m = 2k^2$$

$\therefore n^2$ is even integer , but this is contradiction.

So " if n^2 is odd ,then n is odd"

2.1 SETS:

A set is any well defined collection of objects, called the elements or members of the set.

Examples of sets

1- The collection of all alphabets.
2- The collection of numbers between 0 and 10, etc.

Notes:

1- The elements of set write between braces { }.
2- The order is not important {1, 2, 3} = {3, 1, 2}
3- Repetition is not important {1, 1, 2} = {1, 2}
4- The methods of describing sets by:
   a) listing, such as {1, 2, 3,… }.
   b) special property, such as {x | x is a positive integer less than 4}

Ex: Write the following sets by listing:

a) A= {x | x is a positive integer less than 4 }
b) B= {x | x is a negative integer bigger than -5 }

sol:

a) A= {1, 2, 3}
b) B= {-4,-3,-2,-1}

Ex: Write the following sets by listing:

a) A= {2, 4, 6}
b) B= {1, 4, 9}

sol:

a) A= {x | x is an even integer between 1 and 7 }
b) B= {x | x is the square of integer between 0 and 4 }

Finite Sets

Let S be a set. If there are exactly n elements in S where n is a nonnegative integer, we say that S is a finite set and that n is the cardinality of S. The cardinality of S is denoted by $|S|$.

Ex: What's the cardinality of sets:

a) A={1,2,3}
b) B={-1,-2,-3,-4}

sol:

a) $|A| = 3$
b) $|B| = 4$

Infinite Sets

A set is said to be infinite if it is not finite (i.e., without end, unending).

Some special infinite sets:

1- $\mathbb{N}=\{x \mid x \text{ is a positive integer}\}= \{1, 2,3, \dots\}$, The Natural numbers.

2- $\mathbb{Z}= \{x \mid x \text{ is an integer}\}=\{\dots, -2, -1, 0, 1, 2, \dots\}$, The integers.
$\mathbb{Z}^+ \geq 0= \{x \mid x \text{ is a positive integer}\}=\{ 1, 2, \dots\}$. is the same as $\mathbb{N}$
$\mathbb{Z} \geq 0$ is the set of nonnegative integers.

3- $\mathbb{Q}$ is the set of rational numbers, ratio of integers, 3/6, -17/4, or 3/1.
$\mathbb{Q}^+$ is the set of positive rational.
$\mathbb{Q}\geq 0$ is the set of nonnegative rational.

4- $\mathbb{R} = \{x \mid x \text{ is a real number}\}$. The "Real" numbers, such as
374.18284719294981819172819431 25…
$\mathbb{R}^+$ is the set of positive real number
$\mathbb{R}\geq 0$ is the set of nonnegative real numbers
$\mathbb{R}^* = \mathbb{R}\backslash\{0\}$ is the set of all real numbers except 0

Subset:

The set A is a subset of B if and only if every element of A is also an element of B. We use the notation ($A \subseteq B$). (If A is not a subset of B, we write $A \not\subset B$).

Notes:

1- If A is any set, then $A \subseteq A$ (Every set is a subset of itself).
2- $\varnothing \subseteq A$ for any set A.
3- $A \subseteq B$ and $B \subseteq A$ if and only if $A = B$

Ex: Write $\subseteq$ or $\not\subset$ of the following sets

$\mathbb{Z}^+ \square \mathbb{Z}$? $\mathbb{Z} \square \mathbb{Q}$? A = {1,2,3,4,5,6}, B = {2,4,5}, C = {1,2,3,4,5}

sol:

1) $\mathbb{Z}^+ \subseteq \mathbb{Z}$
2) $\mathbb{Z} \subseteq \mathbb{Q}$
3) $A \not\subset B$
4) $B \subseteq A$
5) $C \subseteq A$
6) $A \not\subset C$
7) $C \not\subset B$
8) $B \subseteq C$

SET OPERATIONS:

1- Union Definition:

Let A and B be sets. The union of the sets A and B, denoted by $(A \cup B)$, is the set that contains those elements that are either in A or B, or in both.

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

Ex: If A = {a, b, c}, and B = {d, e}, then $A \cup B$ = {a, b, c, d,e}

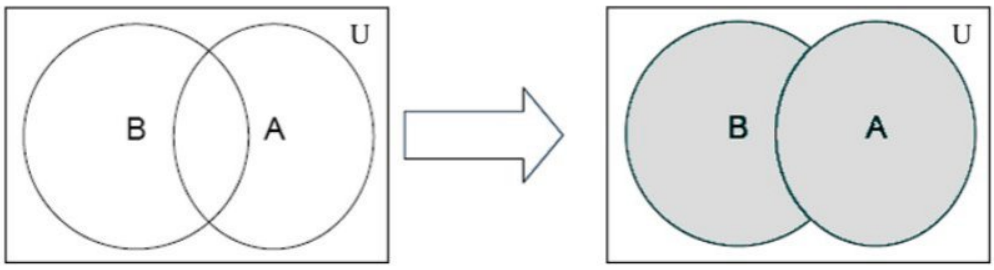


Figure 3: Venn diagram representing the union of A and B (A ∪ B is shaded)

2- Cartesian product of two sets:

Let A and B be sets. The Cartesian of product of A and B is :

$$A \times B = \{(a,b) \mid a \in A \wedge b \in B\}$$

Ex: If A = {a, c, d}, and B = {d, e}, then

$A \times B = \{(a,d),(a,e),(c,d),(c,e),(d,d),(d,e)\}$

3- Intersection

Let A and B be sets. The intersection of the sets A and B, denoted by $(A \cap B)$, is the set that containing those elements in both A and B.

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$

Ex: If A = {a, c, d}, and B = {d, e}, then $A \cap B = \{d\}$

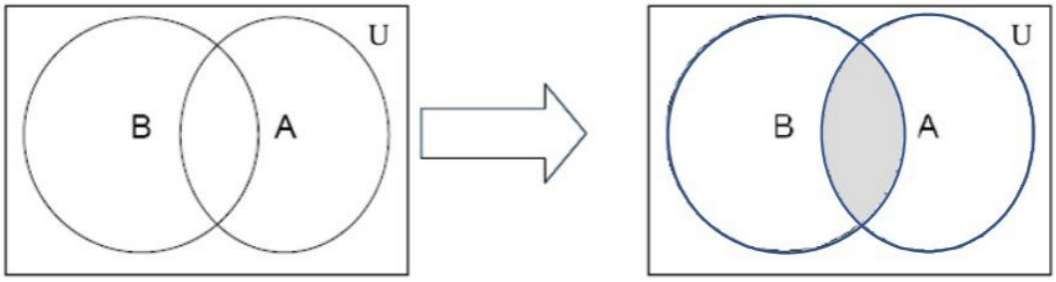


Figure 4: Venn diagram representing the intersection of A and B
(A ∩ B is shaded)

4- Disjoint Sets

Two sets are disjoint if $A \cap B = \emptyset$.

Ex: If A = {a, c, d}, and B = {e}, then A and B are disjoint , because

$$A \cap B = \{\} = \emptyset$$

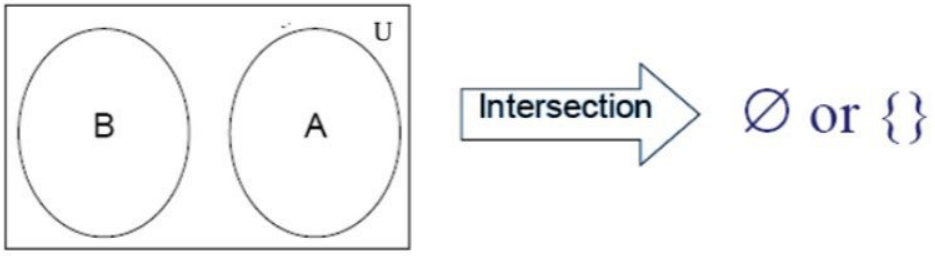


Figure 5: Venn diagram representing the disjoint of A and B

5- Complement set

A Let U be the universal set. The complement of the set A, denoted by $(\bar{A}$ *or* $A^c)$. The complement of the set A is U-A.

$$\bar{A} = \{x : x \notin A\}$$

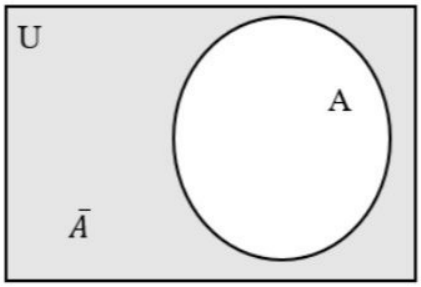


Figure 6: Venn diagram for the complement of the set A

Notes: $\overline{\emptyset} = U$ and $\overline{U} = \emptyset$

Ex: Find the complement of the following sets:

1) A = {x : x is positive integer}
2) B = {1,2} ,where U is all integer numbers

Sol:

1) $\bar{A}$ = {x : x is not positive integer}
2) $\bar{B} = \mathbb{Z}/\{1,2\} = \{\dots, -2, -1, 0, 3, 4, \dots\}$

6- Difference

Let A and B be sets. The difference of A and B, denoted by $(A - B)$, is the set containing those elements that are in A but not in B.

$$A - B = \{x : x \in A \text{ and } x \notin B\}$$

Ex: If A = {a, b, c, d}, and B = {b, e, f}, then $A - B = \{a, c, d\}$

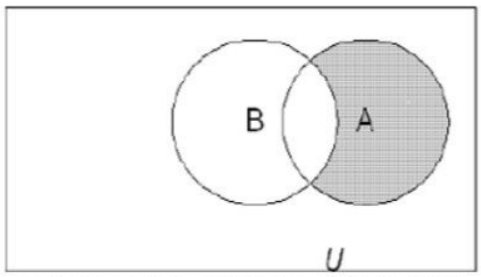


Figure 7: Venn diagram for the difference of A and B (A – B is shaded)

7- Symmetric Difference

The symmetric difference (Like Exclusive OR) , denoted by (A ⊕ B),

$$A \oplus B = (A - B) \cup (B - A)$$
$$= (A \cup B) - (B \cap A)$$

Ex: If A = {a, b, c, d}, and B = {b, c, e}, then $A \oplus B = \{a, d, e\}$

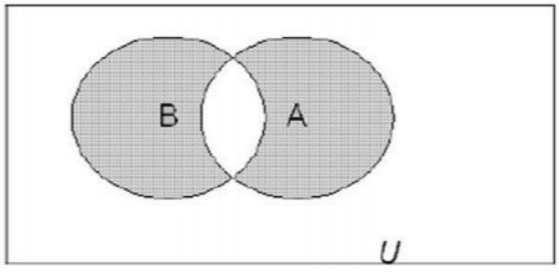


Figure 8: Venn diagram for the symmetric difference of A and B (A ⊕ B is shaded)

$A \cup \emptyset = A$ $A \cap U = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\overline{(\overline{A})} = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
$A \cup (B \cup C) = (A \cup B) \cup C$ $A \cap (B \cap C) = (A \cap B) \cap C$	Associative laws
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	Distributive laws
$\overline{A \cup B} = \overline{A} \cap \overline{B}$ $\overline{A \cap B} = \overline{A} \cup \overline{B}$	De Morgan's laws

Definition:

The union of a collection of sets $A_1, A_2, \dots, A_n$ is the set that contains those elements that are members of at least one set in the collection. Denoted by $\bigcup_{i=1}^{n} A_i$

$$\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \dots \cup A_n$$

Definition:

The intersection of a collection of sets $A_1, A_2, \dots, A_n$ is the set that contains those elements that are members of all the sets in the collection. Denoted by $\bigcap_{i=1}^{n} A_i$

$$\bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap \dots \cap A_n$$

Ex: If $A_i = \{i, i+1, i+2, \dots\}$,find

a) $\bigcup_{i=1}^{n} A_i$ b) $\bigcap_{i=1}^{n} A_i$

Sol:

$$A_1 = \{1,2,3, \dots\}$$

$$A_2 = \{2,3,4, \dots\}$$

$$A_3 = \{3,4,5, \dots\}$$

$$\vdots$$

$$A_n = \{n, n+1, n+2, \dots\}$$

$$\because \quad A_n \subseteq \cdots \subseteq A_3 \subseteq A_2 \subseteq A_1$$

a) $\bigcup_{i=1}^{n} A_i = \{1,2,3, \dots\} = A_1$

b) $\bigcap_{i=1}^{n} A_i = \{n, n+1, n+2, \dots\} = A_n$

2.2 FUNCTIONS:

Definition:

Let A and B be nonempty sets. A function f from A to B $(f : A \to B)$ is an assignment of exactly one element of B to each element of A. We write f(a) = b

Often, we give a formula, such as $f(x) = x + 1$.

Notes:

If f is a function from A to B, s.t $f(a) =$ b

1) A is the domain of f (D_f) and B is the codomain of f.
2) b is the image of a.
3) The range of f $(R_f)$is the set of all images of elements of A.

Ex: If f is a function $(f: \mathbb{Z} \to \mathbb{Z})$,s.t $f(x) = x^2$, find the range of f.

Sol:

$$f(0) = 0^2 = 0$$

$$f(1) = (1)^2 = f(-1) = (-1)^2 = 1$$

$$f(2) = (2)^2 = f(-2) = (-2)^2 = 4$$

$$f(3) = (3)^2 = f(-3) = (-3)^2 = 9$$

$$\vdots$$

$$\therefore \ R_f = \{0, 1, 4, 9, \dots\}$$

Definition:

Let f_1 and f_2 be functions from A to B, then

1) $(f_1 + f_2)$ is also functions from A to B defined by:
$$(f_1 + f_2)(x) = f_1(x) + f_2(x)$$
2) $(f_1 f_2)$ is also functions from A to B defined by:
$$(f_1 f_2)(x) = f_1(x) f_2(x)$$

Definition:

Let f be a function from the set A to the set B $(f : A \to B)$ and let S be a subset of A $(S \subseteq A)$. The image of S under the function f is the subset of B $(f(S) \subseteq B)$. so $f(S) = \{f(s) \mid s \in S\}$.

Ex: Let $A = \{a, b, c, d, e\}$ and $B = \{1, 2, 3, 4\}$ where $f(a) = 2$, $f(b) = 1$, $f(c) = 4, f(d) = 1, f(e) = 1$, find $f(S)$ where $S = \{b, c, d\}$

Sol: $f(S) = \{1, 4\}$

One-to-One and Onto Functions

Definition:

A function f is **one-to-one**, or **injective**, if and only if $\forall a, b \in D_f$

(If $f(a) = f(b)$ then $a = b$) or (If $a \neq b$ then $f(a) \neq f(b)$)

Ex: Determine whether the following functions is one-to-one:

1- A function f from $\{a, b, c, d\}$ to $\{1, 2, 3, 4,5\}$,where $f(a) = 4, f(b) = 5$, $f(c) = 1$, $f(d) = 3$
2- $f: \mathbb{R} \to \mathbb{R}$, s.t $f(x) = x^2$
3- $f: \mathbb{R} \to \mathbb{R}$, s.t $f(x) = x + 1$

Sol:

1- The function f is one-to-one because no two elements in the domain have the same image.

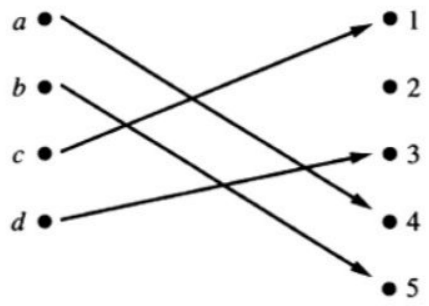


Figure 1: A One-to-One Function.

2- The function $f(x) = x^2$ is not one-to-one because, $1 \neq -1$, but $f(1) = f(-1) = 1$.

Another solution:

Let $a, b \in \mathbb{R}$ and $f(a) = f(b)$

$$\Rightarrow \quad a^2 = b^2 \qquad /\sqrt{\ }$$

$$\Longrightarrow \quad a = \pm b$$

For example $1 \neq -1$, but $f(1) = f(-1) = 1$

$\therefore$ f is one-to-one

3- Let $a, b \in \mathbb{R}$ and $f(a) = f(b)$

$$\Rightarrow \quad a + 1 = b + 1 \qquad /-1$$

$$\Longrightarrow \quad a = b$$

$\therefore$ f is one-to-one

Definition:

A function f from A to B $(f: A \to B)$ is **onto**, or **surjective**, if and only if

$$\forall b \in B\ ,\ \exists a \in A \quad \text{s.t} \quad f(a) = b$$

Ex: Determine whether the following functions is onto:

1- A function f from $\{a, b, c, d\}$ to $\{1, 2, 3\}$,where $f(a) = 3, f(b) = 2$, $f(c) = 1, f(d) = 3$.
2- $f: \mathbb{R} \to \mathbb{R}$, s.t $f(x) = x^2$
3- $f: \mathbb{R} \to \mathbb{R}$, s.t $f(x) = x + 1$

Sol:

1- The function f is onto, because all elements of the codomain are images of elements in the domain, or

$\because$ $R_f = \{1, 2, 3\} =$ Codomain

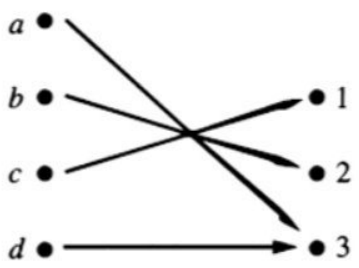


Figure 3: An Onto Function.

2- The function f is not onto because when $y = -1$,then $\nexists x \in \mathbb{R}$ (domain) s.t $x^2 = -1$

Another solution:

Let $b \in \mathbb{R}$ (codomain) s.t $y = x^2 \quad / \sqrt{\ }$

$\Rightarrow \quad x = \pm\sqrt{y} \notin \mathbb{R}$ (domain) ,when $y \in \mathbb{R}^-$

For example $y = -1$,then $\nexists x \in \mathbb{R}$ (domain) s.t $x^2 = -1$

$\therefore$ f is not onto

3- Let $b \in \mathbb{R}$ (codomain) s.t $y = x + 1 \quad /-1$

$\Rightarrow \quad x = y - 1 \in \mathbb{R}$ (domain)

$\therefore$ f is onto

Now we give some examples of correspondences:

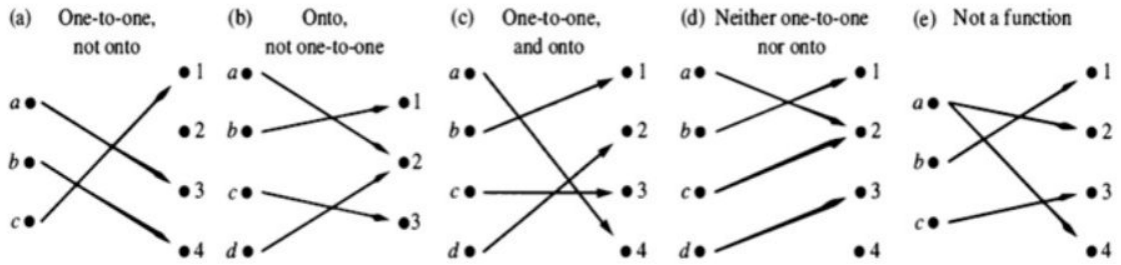


Figure 2: Examples of Different Types of Correspondences.

Definition 8:

The function f is a **one-to-one correspondence**, or a **bijection**, if it is both one-to-one and onto.

Ex: Determine whether the following functions is bijection:

1) A function f from $\{a, b, c, d\}$ to $\{1, 2, 3, 4\}$ with $f(a) = 4, f(b) = 2$, $f(c) = 1, f(d) = 3$.
2) $f: \mathbb{R} \to \mathbb{R}$, s.t $f(x) = x^2$
3) $f: \mathbb{R} \to \mathbb{R}$, s.t $f(x) = x + 1$

Sol:

1) The function f is one-to-one and onto.
   It is one-to-one because no two elements in the domain have the same image.

It is onto because all elements of the codomain are images of elements in the domain or $(R_f = \{1, 2, 3, 4\} =$(codomain)).

$\therefore$ f is a bijection

2) The function f is not one-to-one because $-1 \neq 1$,but $f(-1) = f(1) = 1$

$\therefore$ f is not bijection

3) The function f is bijection because f is one-to-one and onto. (We prove that $f(x) = x + 1$ is one to one and onto in previous examples)

Inverse Functions

Let f be a one-to-one correspondence (bijection). The inverse function of f denoted by (f^{-1}) is an assign each element $b \in B$ to the unique element $a \in A$

$$\text{s.t } f(a) = b \text{ , then } f^{-1}(b) = a.$$

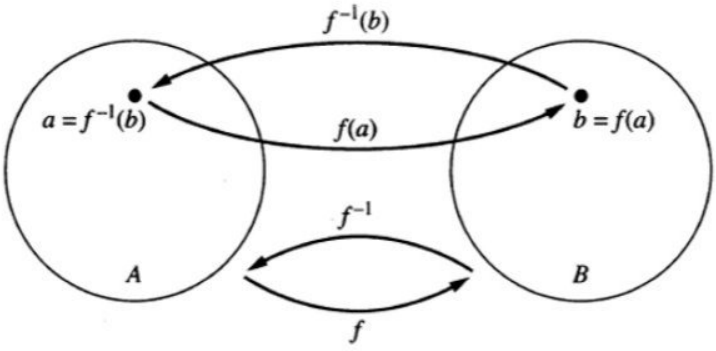


Figure 4: The Function f^{-1} Is the Inverse of Function f.

Ex: Determine whether the following functions is invertible, and if it is, find inverse:

1) A function f from $\{a, b, c\}$ to $\{1, 2, 3\}$ such that $f(a) = 2, f(b) = 3, f(c) = 1$.
2) $f: \mathbb{R} \to \mathbb{R}$, s.t $f(x) = x^2$
3) $f: \mathbb{R} \to \mathbb{R}$, s.t $f(x) = x + 1$

Sol:

1) The function f is invertible because it is a bijection.
   It is one-to-one because no two elements in the domain have the same image.
   It is onto because all elements of the codomain are images of elements in the domain or $(R_f = \{1, 2, 3\} =$(codomain)).

The inverse function of f is $f^{-1}(1) = c,\ f^{-1}(2) = a,\ f^{-1}(3) = b.$

2) The function f is not one-to-one because $-1 \neq 1$,but $f(-1) = f(1) = 1$

$$\therefore\ f \text{ is not invertible}$$

3) The function f is invertible because f is bijection. (We prove that $f(x) = x + 1$ is one to one and onto in previous examples)
The inverse function of f

$$\begin{aligned} y &= x + 1 \quad /-1 \\ \Longrightarrow\ x &= y - 1 \\ \therefore\ f^{-1}(y) &= y - 1 \end{aligned}$$

The Composition Of The Functions:

Let g be a function from the set A to the set B and let f be a function from the set B to the set C. The composition of the functions f and g, denoted by fog, is defined by $(fog)(a) = f(g(a))$.

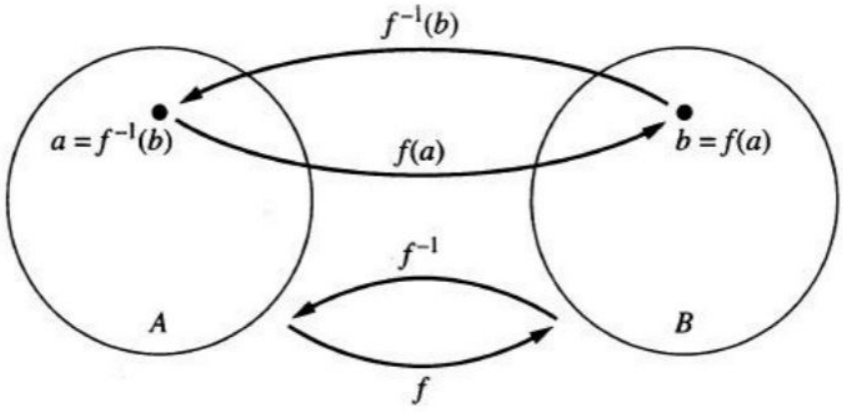


Figure 5: The Composition of the Functions f and g.

Notes:

1) fog is defined if $R_g \subseteq D_f$
2) gof is defined if $R_f \subseteq D_g$

Example 13 :

Let f and g be the functions from $\mathbb{Z}$ to $\mathbb{Z}$,defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$.

Find: 1) fog 2) gof

Solution:

1) $(fog)(x) = f(g(x)) = f(3x + 2) = 2(3x + 2) + 3 = 6x + 7.$
2) $(gof)(x) = g(f(x)) = g(2x + 3) = 3(2x + 3) + 2 = 6x + 11.$

Example 14:

Let g be the function from the set {1, 2, 3} to itself such that g(1)=2, g(2) = 3, and g(3) = 1 .

Let f be the function from the set {1, 2, 3} to the set {4, 5, 6} such that f(1) = 6 , f(2) = 5, and f(3)=4.

Find: 1) fog 2) gof

Solution:

1) The composition fog is defined by:

   (f o g)(a) = f(g(a)) = f(b) = 2

   (f o g) (b) = f(g(b)) = f(c) = 1

   (f o g)(c) = f(g(c)) = f(a) = 3

2) Note that gof is not defined, because $R_f \nsubseteq D_g$.

Some Important Functions

1) The floor function:
The floor function assigns to the real number x the largest integer that is less than or equal to x, denoted by $\lfloor x \rfloor$ or $[x]$.

2) The ceiling function:
The ceiling function assigns to the real number x the smallest integer that is greater than or equal to x, denoted by $\lceil x \rceil$.

Example 16:

Find the value of the following:

1) $\left\lfloor \frac{1}{2} \right\rfloor$ 3) $\lfloor 3.1 \rfloor$ 5) $\left\lceil \frac{1}{2} \right\rceil$ 7) $\lceil 3.1 \rceil$

2) $\left\lfloor -\frac{1}{2} \right\rfloor$ 4) $\lfloor 7 \rfloor$ 6) $\left\lceil -\frac{1}{2} \right\rceil$ 8) $\lceil 7 \rceil$

Solution:

1) $\left\lfloor \frac{1}{2} \right\rfloor = 0$ 3) $\lfloor 3.1 \rfloor = 3$ 5) $\left\lceil \frac{1}{2} \right\rceil = 1$ 7) $\lceil 3.1 \rceil = 4$

2) $\left\lfloor -\frac{1}{2} \right\rfloor = 1$ 4) $\lfloor 7 \rfloor = 7$ 6) $\left\lceil -\frac{1}{2} \right\rceil = 0$ 8) $\lceil 7 \rceil = 7$

SEQUENCES AND SUMMATONS :

Sequences :

Sequences are ordered lists of elements. Usually, mathematical sequences are infinite, denoted by $\{a_n\}_{n=1}^{\infty}$

Example 1 :

Write the elements of the sequence $\{\frac{1}{a}\}_{n=1}^{\infty}$

Solution:

$a_1, a_2, a_3, a_4, \ldots \quad \Longrightarrow \quad 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$

TABLE 1 Some Useful Sequences.

nth Term	*First 10 Terms*
n^2	1, 4, 9, 16, 25, 36, 49, 64, 81, 100, . . .
n^3	1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, . . .
n^4	1, 16, 81, 256, 625, 1296, 2401, 4096, 6561, 10000, . . .
2^n	2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, . . .
3^n	3, 9, 27, 81, 243, 729, 2187, 6561, 19683, 59049, . . .
$n!$	1, 2, 6, 24, 120, 720, 5040, 40320, 362880, 3628800, . . .

Summations

The symbol "∑" takes a sequence of numbers and turns it into a sum .

Symbolically :

$\sum_{i=0}^{n} a_i = a_0 + a_1 + a_2 + \cdots + a_n$

Properties of Summations

1) $\sum_{i=1}^{n}(a_i \pm b_i) = \sum_{i=1}^{n} a_i \pm \sum_{i=1}^{n} b_i$
2) $\sum_{i=1}^{n} c\, a_i = c \sum_{i=1}^{n} a_i$, (c is scalar)
3) $\sum_{i=1}^{n} c = c \times n$, (c is scalar)
4) $\sum_{i=1}^{n} a_i = \sum_{i=1}^{k} a_i + \sum_{i=k+1}^{n} a_i$

Example 2 :

What is the value of the following:

1) $\sum_{j=1}^{5} j^2$ 2) $\sum_{k=4}^{8}(-1)^k$ 3) $\sum_{i=1}^{4}\sum_{j=1}^{3} ij$

Solution:

1) $\sum_{j=1}^{5} j^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2$

$= 1 + 4 + 9 + 16 + 25$

$= 55$

2) $\sum_{k=4}^{8}(-1)^k = (-1)^4 + (-1)^5 + (-1)^6 + (-1)^7 + (-1)^8$

$= 1 + (-1) + 1 + (-1) + 1$

$= 1$

3) $\sum_{i=1}^{4}\sum_{j=1}^{3} ij = \sum_{i=1}^{4}(i + 2i + 3i)$

$= \sum_{i=1}^{4} 6i$

$= 6\sum_{i=1}^{4} i$

$= 6(1 + 2 + 3 + 4)$

$= 6 \times 10$

$= 60$

Countable And Uncountable Sets:

1) **Finite Set:**
   The finite set is called ***countable***.

For examples:

a) $\{1,3,5,7,9\}$ is ***countable***.
b) $\{0,2,4,6,8\}$ is ***countable***.

2) **Infinite Set:**
   When the elements of an infinite set can be listed, the set is called ***countable***. If we can define a bijection function from $\{0,1,2,3, \ldots\}$ to any infinite set, then this set is ***countable***. Otherwise, the set is ***uncountable***

Remark:

1) The set of rational numbers $\mathbb{Q}$ is ***countable***
2) The set of real numbers $\mathbb{R}$ is ***uncountable***.

For examples:

a) $\{1,3,5,7,9, \dots\}$

We can define a bijection function from $\{0,1,2,3, \dots\}$ to $\{1,3,5,7,9, \dots\}$ as follows:

$$f(n) = 2n + 1$$

b) $\{0,2,4,6,8, \dots\}$

We can define a bijection function from $\{0,1,2,3, \dots\}$ to $\{0,2,4,6,8, \dots\}$ as follows:

$$f(n) = 2n$$

Relations and Their Properties

Definition 1 :

Let A and B be sets. A binary relation "R" from A to B is a subset of A×B,

We write $R = \{(a, b): a \in A \wedge b \in B\}$

Example 2 :

Write ordered pairs (elements) of the following relations from A = {0, 1, 2} to B = {2, 3}

1) $R_1 = \{(a, b) : a < b\}$
2) $R_2 = \{(a, b) : a = b\}$

Solution:

$A \times B = \{(0, 2), (0, 3), (1, 2), (1, 3), (2, 2), (2, 3)\}$

1) $R_1 = \{(0, 2), (0, 3), (1, 2), (1, 3), (2, 3)\}$.
2) $R_2 = \{(2, 2)\}$.

Relations on a Set

A relation on the set A is a relation from A to A. In other words, a relation on a set A is a subset o f A x A .

Example 3 :

Write ordered pairs (elements) of the following relations on the set {l, 2, 3, 4}.

1) $R_1 = \{(a, b) : a \text{ divides } b\}$
2) $R_2 = \{(a, b) : a = b + 1\}$

Solution:

1) $R_1 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$.
2) $R_2 = \{(2, 1), (3, 2), (4, 3)\}$.

Exercises :

Write ordered pairs (elements) of the following relations on the set {l, 2, 3, 4}.

1) $R_1 = \{(a, b) : a \leq b\}$
2) $R_2 = \{(a, b) : a > b\}$
3) $R_3 = \{(a, b) : a = b \text{ or } a = -b\}$
4) $R_4 = \{(a, b) : a = b\}$
5) $R_5 = \{(a, b) : a = b + 1\}$
6) $R_6 = \{(a, b) : a + b \leq 3\}$

Properties of Relations

Let "R" is a relation on a set A

1) **Reflexive Relation:**
   If $(a, a) \in R$, $\forall a \in A$
2) **Symmetric Relation:**
   If $(a, b) \in R$ then $(b, a) \in R$, for all $a, b \in A$
3) **Transitive Relation:**
   If $(a, b) \in R$ and $(b, c) \in R$, then $(a , c) \in R$, for all $a , b , c \in A$.

Example 5 :

Determine whether the following relations on $A = \{1, 2, 3, 4\}$ is ***reflexive***, ***symmetric*** or ***transitive***:

- ☒ $R_1 = \{(1, 1), (1 , 2), (2, 1), (2, 2), (3 , 4), (4, 1) , (4, 4)\}$
- ✓ $R_2 = \{(1, 1), (1, 2), (2,1)\}$
- ✓ $R_3 = \{ (1 , 1) , (1 , 2) , (1 , 4), (2, 1), (2, 2), (3 , 3) , (4, 1) , (4, 4)\}$
- ☒ $R_4 = \{(2, 1), (3 , 1) , (3 , 2), (4, 1) , (4, 2), (4, 3)\}$
- ☒ $R_5 = \{(1, 1), (1, 2), (1, 3), (1 , 4), (2, 2), (2, 3), (2, 4), (3 , 3), (3, 4), (4, 4)\}$
- ✓ $R_6 = \{(3, 4)\}$

R_1, R_4 and R_5 exercises

Solution:

✓ $\boldsymbol{R_2} = \{(\mathbf{1}, \mathbf{1}), (\mathbf{1}, \mathbf{2}), (\mathbf{2}, \mathbf{1})\}$

R_2 is not reflexive because $(2, 2) \notin R_2$.

R_2 is symmetric because if $(a, b) \in R_2$,then $(b, a) \in R_2$.

R_2 is not transitive because $(2,1) \in R_2$ and $(1,2) \in R_2$,but $(2, 2) \notin R_2$.

✓ $\boldsymbol{R_3} = \{(\mathbf{1} , \mathbf{1}), (\mathbf{1} , \mathbf{2}) , (\mathbf{1} , \mathbf{4}), (\mathbf{2}, \mathbf{1}), (\mathbf{2}, \mathbf{2}), (\mathbf{3} , \mathbf{3}) , (\mathbf{4}, \mathbf{1}) , (\mathbf{4}, \mathbf{4})\}$

R_3 is reflexive because $(a, a) \in R_3$, $\forall\, a \in A$.

R_3 is symmetric because if $(a, b) \in R_3$,then $(b, a) \in R_3$.

R_3 is not transitive because $(4,1) \in R_3$ and $(1, 2) \in R_3$,but $(4, 2) \notin R_3$.

✓ $\boldsymbol{R_6 = \{(3,4)\}}$

R_6 is not reflexive because $(2,2) \notin R_6$

R_6 is not symmetric because $(3,4) \in R_6$,but $(4,3) \notin R_6$

R_6 is transitive because $(3,4) \in R_6$,but $\nexists (4,c) \in R_6$